

Introduction to Machine Learning

Machine Learning (ML) is a subset of Artificial Intelligence (AI) that enables computers to learn from data and improve their performance without being explicitly programmed. ML systems identify patterns in data and make predictions or decisions automatically.

What is Machine Learning?

Machine Learning allows systems to learn from past experiences and data to make better predictions and decisions.

Types of Machine Learning

- 1. Supervised Learning:** Learning using labeled data.
- 2. Unsupervised Learning:** Finding patterns from unlabeled data.
- 3. Reinforcement Learning:** Learning through rewards and punishments.

Applications of Machine Learning

ML is used in recommendation systems, fraud detection, image recognition, voice assistants, self-driving cars, healthcare diagnosis, and spam filtering.

Advantages of Machine Learning

- Automates tasks
- Improves accuracy over time
- Handles large datasets efficiently
- Helps in data-driven decisions

Disadvantages of Machine Learning

- Requires large amounts of data
- High computational cost
- Risk of biased models
- Difficult to interpret complex models

Popular Machine Learning Algorithms

Linear Regression, Logistic Regression, Decision Trees, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Neural Networks.

Steps in Machine Learning

- 1. Collect Data
- 2. Clean and Prepare Data
- 3. Choose a Model
- 4. Train the Model
- 5. Test the Model
- 6. Deploy the Model

Future of Machine Learning

Machine Learning is shaping the future of technology by enabling automation, smart analytics, predictive systems, and intelligent applications.

Machine Learning Algorithms Overview

Algorithm	Purpose
Linear Regression	Predict continuous values
Decision Tree	Classification and prediction
Random Forest	Improved prediction accuracy
KNN	Classification based on neighbors
Neural Networks	Deep learning applications